

SCM: Sleep-Consolidated Memory for Language Agents

A Reproducible Evaluation of One-Shot Encoding, Sleep-Stage Consolidation, and Selective Forgetting

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Abstract

Large language models remain constrained by memory systems that are either bounded by context windows or implemented as continuously growing retrieval stores without lifecycle management. We present SCM (Sleep-Consolidated Memory), a memory architecture that combines bounded working memory, event-aware encoding, staged sleep consolidation, selective forgetting, and contradiction-safe version lineage. We evaluate SCM using repository-backed benchmark artifacts and reproducibility packs. In the latest human-memory benchmark, SCM attains one-shot recall accuracy 1.0, micro-sleep disambiguation gain 1.0, deep-sleep disambiguation gain 1.0, deep-sleep noise retention 0.0, and contradiction-versioning accuracy 1.0. In a 10-seed baseline stress comparison with 96 memory pairs per seed, lexical retrieval, vector retrieval, and SCM without sleep remain at disambiguation recall 0.0, while SCM with deep sleep reaches mean disambiguation recall 0.9052 with mean noise retention 0.0. In long-horizon evaluation, sleep-enabled SCM reaches final family-aware disambiguation recall 0.9812 and final noise retention 0.0, versus awake-only recall 0.0 and noise retention 1.0. Reproducibility-pack reruns pass end-to-end. These results support SCM as a strong and testable memory-lifecycle architecture and establish a rigorous baseline for external user studies.

1 Introduction

Long-context prompting and retrieval-augmented memory have improved practical agent performance, but they do not fully address long-horizon memory quality under conflict and noise. Context-window approaches degrade when relevant information is not near prompt boundaries [3]. Retrieval stores improve recall but are commonly append-only and weak on consolidation and forgetting [2]. Hierarchical memory systems such as MemGPT improve scale control but are not explicitly modeled as biologically inspired memory lifecycle-

cles [5]. Production memory layers such as Mem0 improve personalization, but still operate as always-awake retrieval systems [1].

Human memory is dynamic and lifecycle-based. It combines bounded working capacity [4], sleep-based consolidation [6], and active forgetting to preserve signal quality [7]. SCM adopts these principles as an engineering architecture for language agents: memory is encoded, replayed, consolidated, selectively forgotten, and contradiction-updated with lineage preservation.

This paper asks a narrow question: does a sleep-centered memory lifecycle produce measurable gains over awake-only and standard retrieval controls under stress testing? We answer using current SCM artifacts and show strong gains in disambiguation and selective noise suppression while preserving one-shot and contradiction-update behaviors.

2 Contributions

This paper makes four concrete contributions:

1. It defines SCM as an end-to-end memory lifecycle with explicit micro-sleep and deep-sleep stages, rather than an always-awake retrieval wrapper.
2. It introduces contradiction-safe version lineage where conflicting updates supersede older traces while preserving auditability.
3. It provides controlled stress evidence showing that sleep stages, not baseline lexical or vector retrieval mechanics, drive disambiguation gains.
4. It ships a reproducibility pathway that replays baseline comparison, long-horizon behavior, guardrails, and smoke stability checks from recorded artifacts.

3 Method

3.1 Architecture Overview

SCM combines bounded working memory with persistent graph-based long-term memory. Incoming dialogue is transformed into typed concepts with contextual tags, then written to working memory and long-term memory with value signals. Retrieval uses semantic cues and spreading activation. Sleep orchestration periodically runs consolidation and forgetting passes, then syncs the updated memory state back into long-term storage.

The architecture is organized into six project phases:

- **Phase 1:** selective encoding and grasp estimation.
- **Phase 2:** event binding and association growth.

- **Phase 3:** spreading-activation retrieval and hypothesis ranking.
- **Phase 4:** micro-sleep and deep-sleep consolidation.
- **Phase 5:** forgetting dynamics and contradiction-safe versioning.
- **Phase 6:** evaluation harnesses, guardrails, and reproducibility pack.

3.2 Sleep Stages

Micro-sleep performs lightweight replay, local reinforcement, and duplicate-pressure cleanup. It targets near-term disambiguation improvements with low disruption.

Deep sleep performs broader replay and consolidation, relation synthesis, and forgetting dynamics. Retired concept states are preserved in sync payloads to maintain contradiction-version lineage through consolidation.

3.3 Selective Forgetting and Version Lineage

Forgetting dynamics compute retention from grasp, salience, rehearsal, association density, recency, and interference. Concepts transition through ACTIVE, SUPPRESSED, and ARCHIVED.

Contradictions are handled through version lineage instead of overwrite semantics: a new conflicting trace can supersede prior versions while preserving root and parent links for traceability.

4 Experimental Protocol

All values reported in this paper are extracted directly from benchmark artifacts:

- Human-memory suite: [research/metrics/phase6_human_memory_latest.json](#)
- Baseline stress comparison: [research/metrics/baseline_comparison_latest.json](#)
- Long-horizon behavior: [research/metrics/long_horizon_latest.json](#)
- Guardrails: [research/metrics/phase6_guardrails_latest.json](#)
- Backend smoke: [research/metrics/phase6_backend_smoke_latest.json](#)
- Reproducibility pack: [research/reproducibility/reproducibility_pack_latest.json](#)

We report metric values and pass flags exactly as stored by these harnesses.

5 Results

5.1 Human-Memory Suite

The latest Phase 6 human-memory artifact reports full pass status. Core metrics are shown in Table 1.

Table 1: Phase 6 human-memory benchmark metrics.

Metric	Value
One-shot recall accuracy	1.0000
Micro-sleep disambiguation gain (absolute)	1.0000
Deep-sleep disambiguation gain (absolute)	1.0000
Deep-sleep noise retention	0.0000
Contradiction-versioning accuracy	1.0000
Overall pass	True

These results indicate that SCM simultaneously preserves one-shot encoding, post-sleep disambiguation, and contradiction-safe update behavior.

5.2 Baseline Stress Comparison

Table 2 summarizes the 10-seed stress comparison with 96 memory pairs per seed. Lexical control, vector control, and SCM awake baseline all stay at disambiguation recall 0.0. Sleep-enabled variants provide large gains.

Table 2: Behavioral stress comparison (10 seeds, 96 pairs per seed).

Setting	Disambiguation Recall Mean
Lexical retrieval baseline	0.0000
Vector retrieval baseline	0.0000
SCM awake-only baseline	0.0000
SCM micro-sleep	1.0000
SCM deep-sleep	0.9052
SCM deep-sleep noise retention mean	0.0000
SCM deep-sleep pass rate	1.0000

The attribution pattern is clear: large gains appear only when sleep consolidation stages are enabled.

5.3 Long-Horizon Behavior

In long-horizon evaluation, the final-day comparison between sleep-enabled and awake-only modes is shown in Table 3.

Table 3: Final long-horizon comparison (day 7).

Metric	Awake-Only	Sleep-Enabled
Family-aware disambiguation recall	0.0000	0.9812
Noise retention	1.0000	0.0000
Anchor hit rate	1.0000	1.0000
Disambiguation lift (sleep - awake)	0.9812	
Noise reduction (awake - sleep)	1.0000	
Strict duplicate-pair lift	0.0000	

The strict duplicate-pair lift remains a transparent stress limitation and is not hidden in reporting.

5.4 Reliability and Guardrails

The guardrail suite reports `overall_pass=True`, with zero hits for tracked deprecation patterns (`datetime.utcnow()` and `.dict()`). The backend smoke artifact reports 20/20 HTTP success, 20/20 overall pass, average latency 45.2 ms, p50 latency 22.7 ms, and p90 latency 26.3 ms.

The reproducibility pack reports `overall_pass=True`, with passing summaries across baseline comparison, long-horizon behavior, guardrails, and smoke pytests.

6 Discussion

Across suites, SCM is strongest where memory quality depends on lifecycle transformations, especially disambiguation under duplicate pressure and selective noise suppression over time. Because awake-only and no-sleep controls remain at or near zero disambiguation in stress settings, the observed gains are unlikely to be explained by retrieval plumbing alone.

The current evidence is internal but rigorous: deterministic harnesses, multi-seed stress tests, regression guardrails, and reproducibility-pack reruns. This supports SCM as an engineering-science result with a clear path to broader external validation.

7 Limitations

This work has four primary limitations.

1. Evaluations are mostly synthetic and harness-driven rather than broad naturalistic user studies.

2. External comparisons remain limited; wider cross-system benchmarks should be added.
3. Strict duplicate-pair behavior under long-horizon pressure remains weaker than family-aware behavior.
4. The paper does not claim consciousness or full human cognition; it evaluates bounded human-like memory dynamics only.

8 Conclusion

SCM demonstrates that a sleep-consolidated memory lifecycle can produce large, reproducible gains over awake-only and standard retrieval controls on targeted memory tasks. The architecture preserves one-shot behavior and contradiction-safe updates while improving long-horizon disambiguation and removing low-value noise. The next step is external user validation followed by broader baseline expansion.

Reproducibility Statement

The benchmark and regression scripts are available in the repository, including:

- [tests/baseline_comparison.py](#)
- [tests/long_horizon_benchmark.py](#)
- [tests/phase6_guardrails.py](#)
- [tests/reproducibility_pack.py](#)

The latest reproducibility summary is stored at [research/reproducibility/reproducibility_pack_latest.json](#), which reports `overall_pass=True` as of May 1, 2026.

References

- [1] Taranjeet Singh Chhikara et al. Mem0: Personalized AI memory layer. *arXiv preprint arXiv:2504.19413*, 2025.
- [2] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

- [3] Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *arXiv preprint arXiv:2307.03172*, 2023.
- [4] George A Miller. The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review*, 63(2):81–97, 1956.
- [5] Charles Packer, Vivian Fang, Shishir G Patil, Kevin Lin, Joseph Wooders, and Joseph E Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.
- [6] Björn Rasch and Jan Born. About sleep’s role in memory. *Physiological Reviews*, 93(2):681–766, 2013.
- [7] Giulio Tononi and Chiara Cirelli. Sleep and synaptic homeostasis: A hypothesis. *Brain Research Bulletin*, 62(2):143–150, 2003.